

Industrial hygiene assessment of reticuloendotheliosis viruses exposure in the poultry industry.

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OBJECTIVES: Reticuloendotheliosis viruses (REV) are a group of retroviruses like avian leukosis/sarcoma viruses (ALSV) that naturally infect and cause cancers in chickens. We recently found that ALSV antibody levels were associated with job tasks in the poultry industry. The objectives of this study are to examine whether a similar association can be found with REV antibody levels and to examine the correlation between REV and ALSV antibody levels. METHODS: Relative risk was estimated comparing REV antibody levels of 45 poultry workers with those of 44 controls. The expected mean antibody level was predicted for the association with employment by a generalized linear model. Correlation coefficient was measured between ALSV and REV antibody levels. RESULTS: REV antibody levels were significantly higher in poultry workers than in control subjects and were associated with gender and employment conditions, especially employment duration. The relative risk was significantly higher for some job categories. A significant correlation was observed between REV and ALSV antibody levels, which was strong among poultry workers, but weak among the control subjects. CONCLUSION: Antibody levels can be validly used to identify certain job tasks associated with high risk of exposure to REV in the workplace, and the practical implication is recommendations for protection at these job tasks. Importantly, in situations where there is exposure to multiple pathogens in the workplace, the analysis of antibody levels of one pathogen may sufficiently represent exposure to the other correlated pathogens. This suggested exposure assessment may hold true for pathogens with a similar route of transmission.